

1

CIA Triad

Confidentiality: Confidentiality means that only the authorized people are allowed access to the information. This means that people who don't have access to the specific information are not allowed to view, edit or modify it and this thus protects the user specific information or data.

Integrity: Integrity means that the Data is safe and secure and not tempered with. In simple words we can say that it is a way to ensure that the data or information you thinks is yours is in reality yours.

Example: If you have jewels in the bank then integrity means they are actually your jewels not just some stones or other jewellery made of glass.

Availability: It means that the authorized user can access his information any time without any trouble.

2

Virus vs Ransomware

Virus: Virus is a malicious software that is designed to harm your computer. It steals, alters or deletes information. They attach to legitimate documents and then spread to other systems. They require user action to spread.

Ransomware: Ransomware is a malicious program that encrypts the user's drive and then the owner of the ransomware (the bad guy) demands ransom to unencrypt the drive.

3

Suspicious email

If we get a suspicious email from "CEO@yourcompany.com" asking for gift cards then we should not open any attached files in the email, rather we should mark it as spam. For further verification, we should contact the IT Admin about these spam emails and also ask the CEO if he sent these emails. If we want to study these emails then we should open them in sandbox.

4

Authentication Factors

Knowledge: Knowledge or **Something you Know** is information that is known only to you such as the passwords, pins security questions etc.

Possession: Possession or **Something you have** is a physical device such as USB, your mobile phone or any other physical device like smart cards which can be used as key to gain access to your system.

Inherence: Inherence or **Something you are** is something that you were born with like you retina scan or your fingerprints, voice recognition. They are unique only to you.

5

Outdated Software

Using outdated software poses the risk of flaws or bugs in the software that the bad guy or hacker can exploit and gain access to sensitive information which would violate the CIA. The outdated software no longer receive security patches against known vulnerabilities and so the hacker can exploit those flaws.